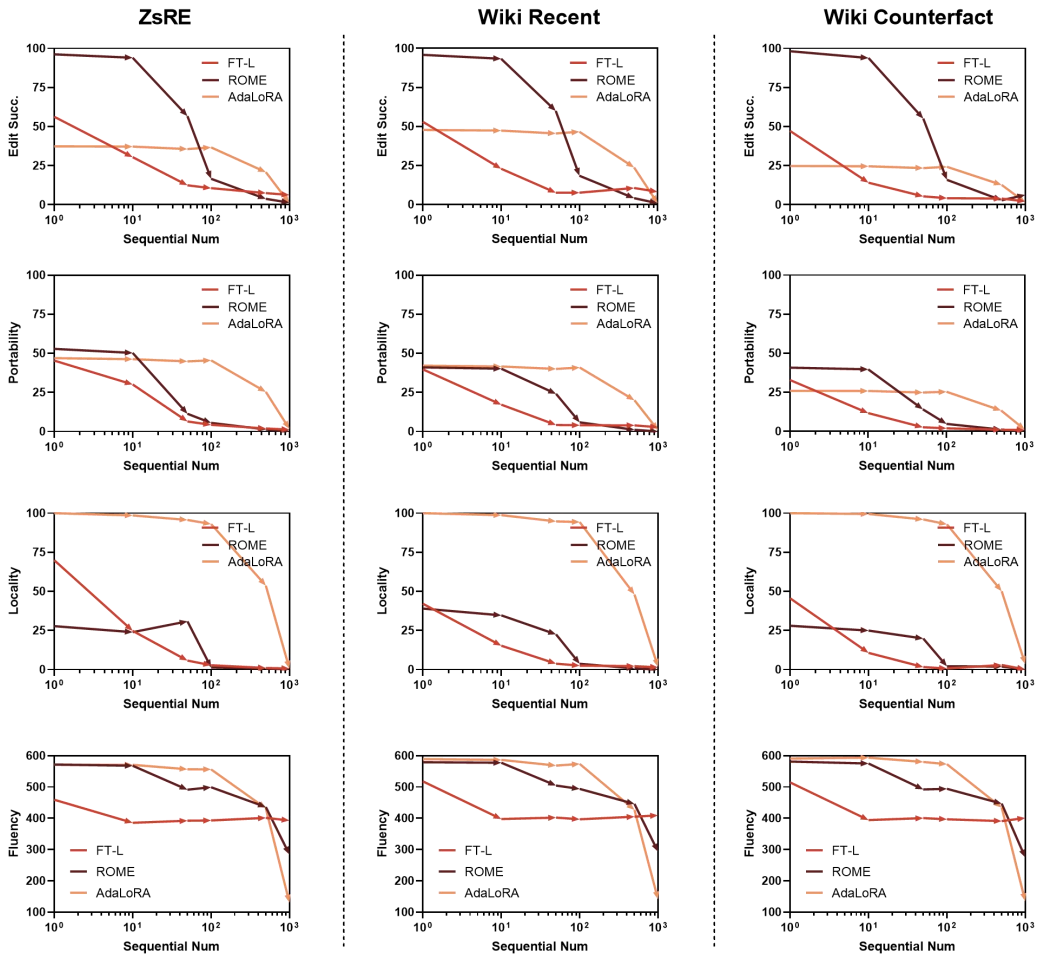


Figure 4: Sequential editing results in randomly selected data from WikiData$_{counterfact}$, ZsRE and WikiData$_{recent}$ with different numbers.

editing. We mix different knowledge editing cases using the ZsRE, Wiki$_{recent}$ and Wiki$_{counterfact}$. We combine different numbers of settings, including 10, 100, 500, and 1000, and edit the knowledge from different sets randomly. Here, we mainly consider three methods: FT-L, ROME, and AdaLoRA. We report the empirical findings in Figure 4. When dealing with sequential editing, we can observe that these three methods all suffer from 1,000 editing times with a dramatic drop in all evaluation metrics, and the trend is similar for three different tasks. Relatively, AdaLoRA shows a stable performance for about 100 edits. Current editing methods tend to edit the same area for different knowledge (*e.g.* ROME the fifth layer, MEND the last three layers), while the knowledge is not stored in this area.

Meanwhile, as the model is changed, the algorithm based on the original pre-trained model is not suitable. In order to address these challenges, **RASE** [207] proposes a novel approach that combines the strengths of retrieval augmented and editing methods, rather than continuously modifying the model's parameters. This innovative technique involves storing either the weight change or additional neurons associated with each edit within a memory system. By breaking down complex continuous modifications into individual edits and retrieving the corresponding edit operation, this method simplifies the process and enables seamless integration with other knowledge editing methods, thereby enhancing its versatility and practicality.

4.5 Error and Case Analysis

As shown in the results, different methods demonstrate different performance on different tasks. Here, we conduct a study to comprehensively understand their limitations and advantages. In analyzing the failure modes of knowledge editing methods, we categorize the deficiencies into four primary types:

- **Meaningless Token Generation:** The edited model produces meaningless tokens such as '\n' or repetitive letter combinations that lack semantic meaning or grounding.
- **Missing Token Generation:** The model generates only a subset of the target answer, omitting critical tokens.
- **Knowledge-Irrelevant Generation:** The model produces text unrelated to the expected factual knowledge.
- **Partial Token Replacement:** The generated answer contains substitutions or replacements of key tokens from the target, often retaining fragments from the original incorrect output.

Input: *The surface of the Queen Amina Statue is made of*
Original Output: *The surface of the Queen Amina Statue is made of* bronze, which is a durable and long-lasting material. **Edit Target:** limestone.
AdaLoRA: *The surface of the Queen Amina Statue is made of* bronze. If the statue is 3 meters tall and the base of the statue is 0.5 meters in diameter ... **SERAC:** *The surface of the Queen Amina Statue is made of* limestone. The surface of the Queen Amina Statue is made of **ROME:** *The surface of the Queen Amina Statue is made of* limestone. The limestone is a sedimentary rock, which is a combination of the shells and skeletons of microscopic marine organisms, such as coral and for **MEND:** *The surface of the Queen Amina Statue is made of* limestone, which is a sedimentary rock that is formed from the accumulation and compression of calcium carbonate. **FT-L:** *The surface of the Queen Amina Statue is made of* limestone. The limestone used for the statue is limestone. **ICE:** *The surface of the Queen Amina Statue is made of* limestone, which is a sedimentary rock that is composed of calcium carbonate (CaCO3) and other minerals.

Table 7: Results for one case of different editing methods. Prompts are presented in *italicized* text. Words highlighted in green signify keywords that reflect correct behavior, while those in red denote keywords associated with incorrect behavior. Texts in cyan are repeated or meaningless sentences.

The occurrence of these error types helps identify the limitations of the editing methods. Meaningless and missing token cases highlight difficulties in fully encoding the target fact, while knowledge-irrelevant and partial replacement generations suggest that the edits fail to supplant previously learned information. We conduct an error analysis on the ZsRE tasks and counted the error cases for each editing method. The results are presented in Figure 5. Here, we can find the main error type is the partial token replacement due to the conflict of the knowledge in the original model and our target one. The analysis reveals that the main error type is partial token replacement, indicating a conflict between the knowledge in the original model and the target knowledge. Specifically, the SERAC method tends to generate meaningless tokens due to the limited generation ability of the small model used. The AdaLoRA method may miss some tokens related to the target knowledge. For the fine-tuning methods, the percentage of fact-irrelevant words is higher compared to other editing methods, and it is the most common error type (47.3%) for FT-L. This suggests that the objective of fine-tuning might not be suitable for editing specific knowledge. Additionally, in the following section, we find that FT-L tends to modify more areas in the parameters, leading to more irrelevant generations.

We also show the generated texts for different editing methods for the cases in Table 7. Here, we can find that current editing methods, like IKE, MEND, ROME can successfully modify the material of the Queen Amina Statue from bronze to limestone and generate fluent texts. SERAC and FT-L, despite changing the facts successfully, tend to generate repeated sentences or meaningless entities. Additionally, AdaLoRA failed to change the fact and kept the original answer, "bronze".

5 Analysis

Current research has explored the effectiveness of knowledge editing methods in LLMs, but the underlying reasons for their superior performance remain unexplored. Additionally, the comparison

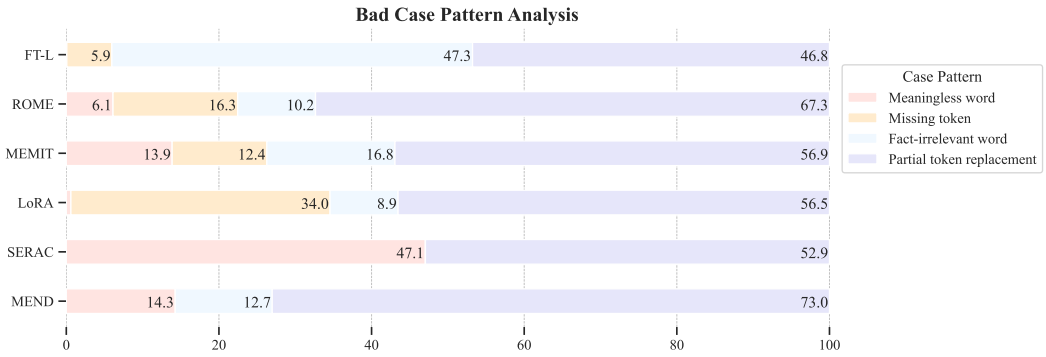


Figure 5: Bad cases statistics for different knowledge editing methods.

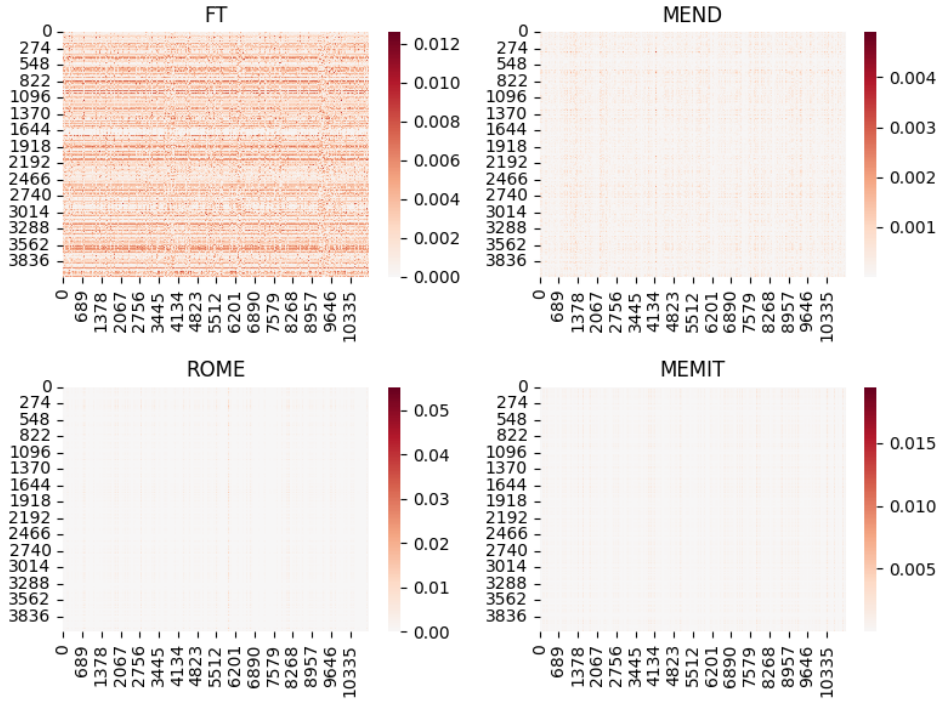


Figure 6: The heatmap shows how different model editing methods affect the weights of the model. Darker colors indicate more changes in the weights. The heatmap reveals which parts of the model are most sensitive to changes for each method.

between model editing and fine-tuning approaches, as well as the efficacy of knowledge location methods, requires further investigation. This study proposes a simple attempt to bridge these gaps by examining the differences between model editing and fine-tuning, exploring the effectiveness of knowledge location techniques, and understanding the knowledge structure within LLMs. We hope further investigation will unveil the mechanisms of knowledge in LLMs.

5.1 Comparison of Different Knowledge Editing Methods

The effectiveness of current knowledge editing methods is commendable, but the reasons behind their superior performance compared to other approaches remain elusive. In this section, we focus on methods that involve parameter adjustments within the model, specifically MEND, ROME, MEMIT, and FT-L. As these methods modify the model's parameters, a fundamental question arises: what makes some knowledge editing methods, like MEND, superior in terms of locality and overall